Advanced Mathematical Techniques Worksheet 7
Dr. Osborne 2021

The purpose of this sheet is to give you some more practice with using regulators and expansions to determine the value of integrals – especially those that push through standard limits on our integral representations – and some practice working with the Riemann zeta function. Please show all of your work and check your answers in Mathematica.

Determine the value of the given integral. Simplify your result as much as possible, leaving answers in terms of π, the Gamma function evaluated at a number between 0 and $1/2$, and the zeta function evaluated at odd numbers. Do not give a value for integrals that do not converge. If an integral does not converge, simply say why and move on.

1) $\int_0^1 \frac{\ln^2\left(1-t^2\right)}{t^4}\,dt$.

2) $\int_0^1 \frac{\ln^2\left(1-t\right)}{t^3\sqrt{t}}\,dt$.

3) $\int_0^1 \frac{\ln^2\left(1-t^2\right)\ln t}{t^3}\,dt$.

4) $\int_0^\infty \frac{1-e^{-3x}}{x\sqrt{x}}\,dx$.

5) $\int_0^\infty \frac{x}{e^{3x}-1}\,dx$.

6) $\int_0^\infty \frac{x^2\,e^{2x}}{\left(e^{2x}-1\right)^2}\,dx$.

7) $\int_0^\infty \frac{x^3}{\left(e^{2x}-1\right)^2}\,dx$.

8) $\int_0^1 \frac{\ln^2(1-t)\ln t}{t^2}\,dt$.